

Feynman's Trick and Generator of the Gamma Function

Ethan Huecker

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The essence of Feynman's integration trick, or "differentiating under the integral sign", is best explained through an example. Let's say we are to determine the value of the integral

$$I = \int_0^1 \frac{x^2 - 1}{\ln(x)} dx, \quad (1)$$

which is difficult due to the $\ln(x)$ in the denominator. Feynman's trick is to remember that $\frac{\partial}{\partial t} x^t = \ln(x)x^t$, and to replace the original integral as a function of t ,

$$I(t) = \int_0^1 \frac{x^t - 1}{\ln(x)} dx, \quad (2)$$

where we agree that $I(2) = I$ is the integral we are concerned with. Now, we will differentiate (2) wrt. this new variable t ,

$$\begin{aligned} \frac{dI(t)}{dt} &= \frac{d}{dt} \int_0^1 \frac{x^t - 1}{\ln(x)} dx = \int_0^1 \frac{\partial}{\partial t} \left(\frac{x^t - 1}{\ln(x)} \right) dx = \int_0^1 x^t dx \\ &= \frac{1}{t+1}, \end{aligned} \quad (3)$$

where clearly the logarithms canceled and I used the fact that $1^{t+1} = 1 \forall t$. We need to get back to (1), so this differential equation must be solved for $I(2)$,

$$I(2) = \int_0^2 \frac{dt}{t+1} = \ln(3) - \ln(1) = \ln(3) \implies \boxed{\int_0^1 \frac{x^2 - 1}{\ln(x)} dx = \ln(3)} \quad (4)$$

In my opinion, Feynman's Trick requires a decent amount of intuition, but with enough experience it can become second-nature. Let's do one more example, namely, the famous "complex analysis" integral

$$I = \int_{-\infty}^{\infty} \frac{\sin(x)}{x} dx. \quad (5)$$

Instead, note that since $\frac{\partial}{\partial t} e^{-tx} = -xe^{-tx}$, if I instead consider the following integral

$$I(t) = 2 \int_0^{\infty} \frac{\sin(x)e^{-tx}}{x} dx, \quad (6)$$

where my goal is to solve for the $I(0)$ case¹, then I can kill the x in the denominator through differentiation:

$$\begin{aligned} \frac{dI(t)}{dt} &= -2 \int_0^{\infty} e^{-tx} \sin(x) dx = -\frac{1}{i} \int_0^{\infty} e^{(i-t)x} - e^{-(i+t)x} dx = -\frac{1}{i} \left[\frac{e^{(i-t)x}}{i-t} + \frac{e^{-(i+t)x}}{i+t} \right]_0^{\infty} \\ &= \frac{1}{i} \left[\frac{1}{i-t} + \frac{1}{i+t} \right] \\ &= -\frac{1}{i} \frac{2i}{1+t^2} \\ &= -\frac{2}{1+t^2}. \end{aligned} \quad (7)$$

¹Since $\sin(x)/x$ is an even function, I can integrate from $0 \rightarrow \infty$ and double it to recover (5), after setting $t = 0$ of course.

This is a differential equation readily solvable by an inverse-tangent, giving

$$I(t) = C - 2 \tan^{-1}(t). \quad (8)$$

In order to fix the constant, note that as $t \rightarrow \infty$, $I(t) \rightarrow 0$ in (6). Therefore, since the horizontal asymptote of $\tan^{-1}(t)$ is $\pi/2$, we must have $C = \pi$ such that

$$I(t) = \pi - 2 \tan^{-1}(t) \implies I(0) = \int_{-\infty}^{\infty} \frac{\sin(x)}{x} dx = \pi \quad (9)$$

Generator of the Gamma Function

Let's look at a seemingly unrelated integral,

$$\int_0^{\infty} e^{-tx} dx = \frac{1}{t}. \quad (10)$$

It is simple to show that this integral is the generator of the gamma function $\Gamma(x)$ under repeated differentiation. Let's look at subsequent derivatives of (10):

$$\begin{aligned} \frac{d}{dt} &\implies -\int_0^{\infty} x e^{-tx} dx = -\frac{1}{t^2} \\ \frac{d^2}{dt^2} &\implies \int_0^{\infty} x^2 e^{-tx} dx = \frac{2}{t^3} \\ &\dots \\ \frac{d^n}{dt^n} &\implies \int_0^{\infty} x^n e^{-tx} dx = \frac{n!}{t^{n+1}} \end{aligned} \quad (11)$$

Setting $t = 1$ in the final integral gives the gamma function $\Gamma(n + 1)$:

$$n! = \int_0^{\infty} x^n e^{-x} dx \equiv \Gamma(n + 1) \quad (12)$$